



Service Tool Instruction

Service Tool Instruction
Number: 3377830-2

Revision Level: 2

Released Date: 25-apr-
2002

Description

Fuel Injector Sleeve Extractor Kit

Purpose

The fuel injector sleeve extractor kit, Part Number 3163523, is used to remove the fuel injector sleeves from the L10, M11, ISM, NH, NT, and N14 engines. This extractor can be used with the cylinder head installed on the engine when used with other tools in the fuel injector sleeve replacement kit, Part Number 3163113. (Note that Part Number 3163113 is **not** available for purchase.) Refer to Service Bulletin 3666324 for procedures to replace the sleeves with the cylinder head installed on the engine. This extractor also can be used for cylinder head rebuild. Refer to the appropriate shop manual for procedures and other tools required to replace the sleeves with the cylinder head removed from the engine.

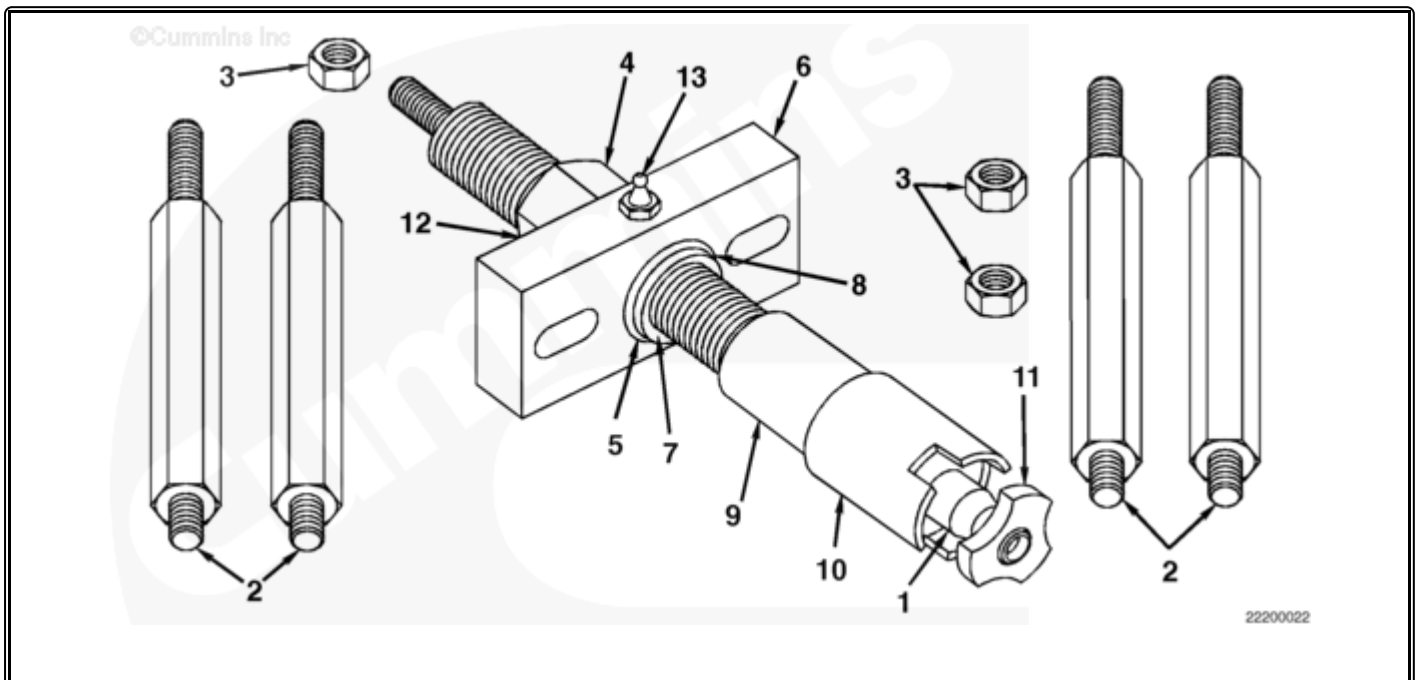


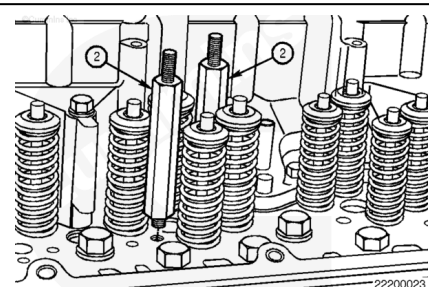
Table 1, Fuel Injector Sleeve Extractor Kit, Part Number 3163523

Item Number	Part Number	Description	Quantity
1	3163637	Extractor adapter	1
2	3163518	Mounting adapter, L10/M11	2

2	3163517	Mounting adapter, N14	2
3	3163519	Hex-flanged nut	5
4	3163509	Forcing nut	1
5	3163506	Thrust washer	1
6	3163510	Remover plate	1
7	3163520	Retaining ring	1
8	3163636	Square key (not shown)	1
9	3163508	Forcing screw	1
10	3163507	Forming collar	1
11	3163505	Extractor	1
12	3163634	Thrust washer	1
13	3163635	Grease fitting	1
Not shown	3163522	Socket setscrew (extractor)	1
Not shown	3163633	Socket setscrew (forming collar)	1

⚠ CAUTION ⚠

Do not use an impact tool for any part of this procedure. Doing so can damage the tool and/or the engine and void the manufacturer's warranty.



Note : There are two sets of clamping supports: One set is for the L10 and M11 engines, and one is for the N14 engine. The shorter threaded end mounts to the engine head.

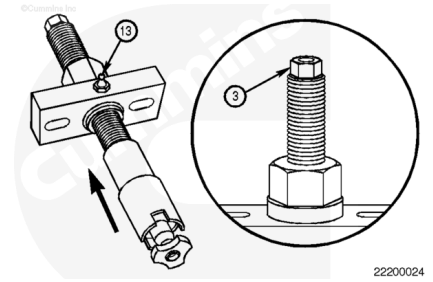
Install the mounting adapters (2) for the appropriate engine.

Tighten the mounting adapters (2).

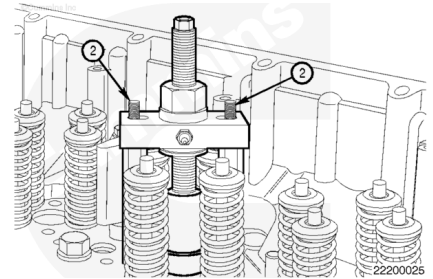
Torque Value: 34 n·m [25 ft-lb]

Use the grease fitting (13) to lubricate the tool until the excess grease extrudes from the thrust washers. Wipe the excess grease onto the threads.

On the sleeve extractor assembly, thread the shaft up into the large nut on the bridge assembly. Loosen the small hex-flanged nut (3) on the through shaft so that the center assembly can move freely up and down inside the shaft.

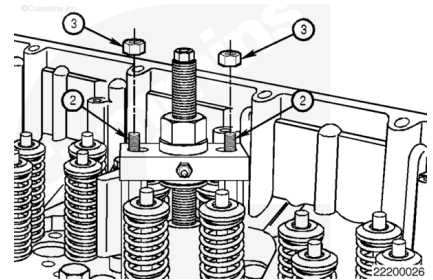


Install the sleeve extractor assembly into the injector bore and onto the mounting adapter (2).



Install the hex-flanged nuts (3) on the mounting adapter (2).
Tighten the hex-flanged nuts.

Torque Value: 34 N·m [25 ft-lb]

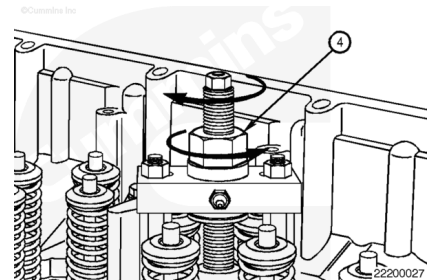


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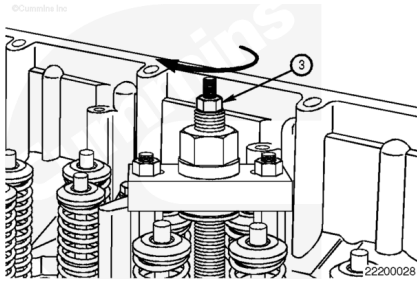
Rotate the shaft **clockwise** or the forcing nut (4) **counterclockwise** until the tool makes contact with the copper sleeve. Continue to rotate the forcing nut (4) **counterclockwise** an additional 5 to 5½ turns until the tool bottoms out against the copper sleeve.

Do **not** exceed a torque of 102 N·m [75 ft-lb].



Tighten the small shaft hex-flanged nut (3) at the top.

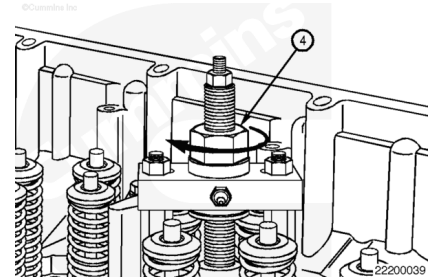
Torque Value: 40 N·m [30 ft-lb]



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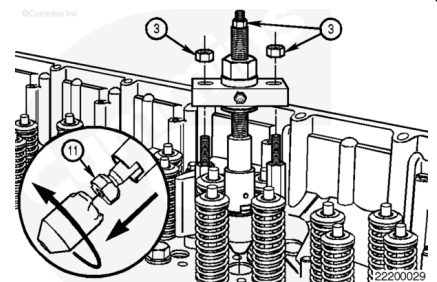
Rotate the forcing nut (4) **clockwise** to remove the copper sleeve from the injector bore.



Loosen the two hex-flanged nuts (3), and remove the sleeve extractor assembly. Loosen the hex-flanged nut (3) at the top of the sleeve extractor assembly shaft.

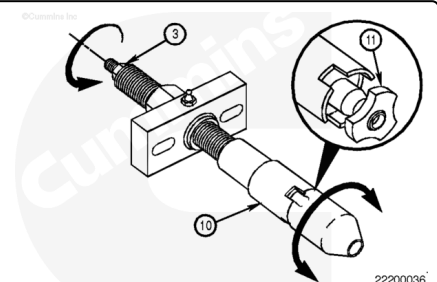
Push the center shaft into the tool, and tap on the copper sleeve to loosen. Rotate the copper sleeve to align the extractor (11) so that the copper sleeve can be removed from the tool.

Note : If the copper sleeve can not be removed by hand, proceed to the next step.



Rotate the copper sleeve to align the extractor (11) and forming collar (10) teeth with the undeformed areas of the copper sleeve.

Tighten the hex-flanged nut (3) until the copper sleeve is forced off the extractor (11).



Remove the mounting adapter (2).

